Group-207
VLSM TREE

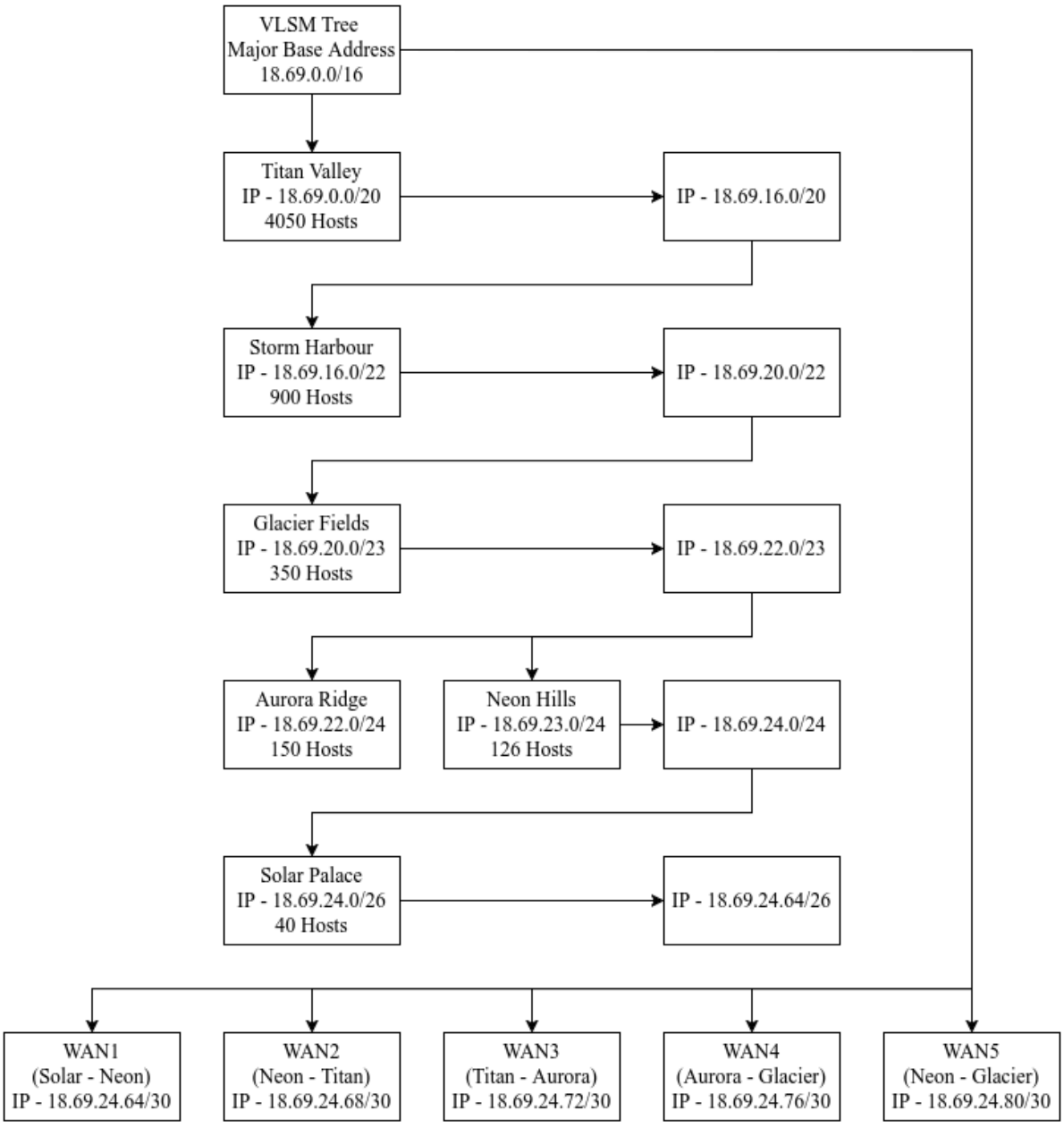


Figure 1 - VLSM Tree Diagram for The Crystal Empire Network

Major Network: 18.69.0.0/16
Available IP addresses in major network: 6208

Number of IP addresses needed: 5642

Subnet Name	Needed Size	Allocated Size	Address	Mask	Dec Mask	Assignable Range
Titan Valley	4052	4094	18.69.0.0	/20	255.255.240.0	18.69.0.1 - 18.69.15.254
Storm Harbour	902	1022	18.69.16.0	/22	255.255.252.0	18.69.16.1 - 18.69.19.254
Glacier Fields	352	510	18.69.20.0	/23	255.255.254.0	18.69.20.1 - 18.69.21.254
Aurora Ridge	152	254	18.69.22.0	/24	255.255.255.0	18.69.22.1 - 18.69.22.254
Neon Hills	132	254	18.69.23.0	/24	255.255.255.0	18.69.23.1 - 18.69.23.254
Solar Palace	42	62	18.69.24.0	/26	255.255.255.192	18.69.24.1 - 18.69.24.62
WAN1 (Solar - Neon)	2	2	18.69.24.64	/30	255.255.255.252	18.69.24.65 - 18.69.24.66
WAN2 (Neon - Titan)	2	2	18.69.24.68	/30	255.255.255.252	18.69.24.69 - 18.69.24.70
WAN3 (Titan - Aurora)	2	2	18.69.24.72	/30	255.255.255.252	18.69.24.73 - 18.69.24.74
WAN4 (Aurora - Glacier)	2	2	18.69.30.0	/30	255.255.255.252	18.69.30.1 - 18.69.30.2
WAN5 (Neon - Glacier)	2	2	18.69.30.4	/30	255.255.255.252	18.69.30.5 - 18.69.30.6

IP address Table

Devices	Interface	IPV4 Address	Subnet mask	Default gateway
Solar Palace	Fa0/0	18.69.24.1	255.255.255.192	N/A
	Se0/0/0	18.69.24.65	255.255.255.252	N/A
Storm Harbour	Fa0/0	18.69.16.1	255.255.252.0	N/A
Titan Valley	Fa0/0	18.69.0.1	255.255.240.0	N/A
	Se0/1/0	18.69.24.70	255.255.255.252	N/A
	Se0/0/1	18.69.24.74	255.255.255.252	N/A
				N/A
Aurora Ridge	Fa0/0	18.69.22.1	255.255.255.0	N/A
	Se0/0/0	18.69.30.1	255.255.255.252	N/A
	Se0/0/1	18.69.24.73	255.255.255.252	N/A
				N/A
Neon Hills	Fa0/0	18.69.23.1	255.255.255.0	N/A
	Se0/0/0	18.69.24.66	255.255.255.252	N/A
	Se0/0/1	18.69.24.69	255.255.255.252	N/A
	Se0/1/0	18.69.30.5	255.255.255.252	N/A
Glacier Fields	Fa0/1	18.69.20.1	255.255.254.0	N/A
	Se0/0/0	18.69.30.6	255.255.255.252	N/A
	Se0/0/1	18.69.30.2	255.255.255.252	N/A
PC3	Fa0/2	18.69.24.10	255.255.255.192	18.69.24.1
PC4	Fa0/3	18.69.24.11	255.255.255.192	18.69.24.1
PC5	Fa0/4	18.69.24.12	255.255.255.192	18.69.24.1
PC3(2)	Fa0/2	18.69.23.20	255.255.255.0	18.69.23.1
PC4(2)	Fa0/4	18.69.23.21	255.255.255.0	18.69.23.1
PC5(2)	Fa0/3	18.69.23.22	255.255.255.0	18.69.23.1
PC3(1)	Fa0/2	18.69.16.20	255.255.252.0	18.69.16.1
PC4(1)	Fa0/4	18.69.16.21	255.255.252.0	18.69.16.1

PC5(1)	Fa0/3	18.69.16.22	255.255.252.0	18.69.16.1
PC3(1)(1)	Fa0/3	18.69.0.10	255.255.240.0	18.69.0.1
PC4(1)(1)	Fa0/2	18.69.0.11	255.255.240.0	18.69.0.1
PC5(1)(1)	Fa0/1	18.69.0.12	255.255.240.0	18.69.0.1
PC3(2)(1)	Fa0/2	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC4(2)(1)	Fa0/4	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC5(2)(1)	Fa0/3	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC3(2)(2)	Fa0/2	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC4(2)(2)	Fa0/4	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC5(2)(2)	Fa0/3	DHCP Assigned	DHCP Assigned	DHCP Assigned
DHCP server	Fa0/8	18.69.23.10	255.255.255.0	18.69.23.1
Web Server	Fa0/6	18.69.23.3	255.255.255.0	18.69.23.1
DNS Server	Fa0/7	18.69.23.2	255.255.255.0	18.69.23.1
Email Server	Fa0/5	18.69.23.5	255.255.255.0	18.69.23.1

Configuration Commands:

Assigning IP address:

For Titan Valley

```
Titan_Valley>en
Titan_Valley#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Titan_Valley(config)#Hostname Titan_Valley
Titan_Valley(config)#interface f0/0
Titan_Valley(config-if)#ip address 18.69.0.1 255.255.240.0
Titan_Valley(config-if)#no shutdown
Titan_Valley(config-if)#exit
Titan_Valley(config)#exit
```

For Storm Harbour

```
Storm_Harbour>en
Storm_Harbour#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Storm_Harbour(config)#hostname Storm_Harbour
Storm_Harbour(config)#interface f0/0
Storm_Harbour(config-if)#ip address 18.69.16.1 255.255.252.0
Storm_Harbour(config-if)#no shutdown
Storm_Harbour(config-if)#exit
Storm_Harbour(config)#exit
Storm_Harbour#
```

For Glacier Fields

```
Glacier_Fields>en
Glacier_Fields#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Glacier_Fields(config)#hostname Glacier_Fields
Glacier_Fields(config)#interface f0/1
Glacier_Fields(config-if)#ip address 18.69.20.1 255.255.254.0
Glacier_Fields(config-if)#no shutdown
Glacier_Fields(config-if)#exit
```

```
Glacier_Fields(config)#exit
Glacier_Fields#
```

For Aurora Ridge

```
Aurora_Ridge>en
Aurora_Ridge#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Aurora_Ridge(config)#hostname Aurora_Ridge
Aurora_Ridge(config)#interface f0/0
Aurora_Ridge(config-if)#ip address 18.69.22.1 255.255.255.0
Aurora_Ridge(config-if)#no shut
Aurora_Ridge(config-if)#exit
Aurora_Ridge(config)#exit
```

For Neon Hills

```
Neon_Hills>en
Neon_Hills#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Neon_Hills(config)#hostname Neon_Hills
Neon_Hills(config)#interface f0/0
Neon_Hills(config-if)#ip address 18.69.23.1 255.255.255.0
Neon_Hills(config-if)#no shut
Neon_Hills(config-if)#exit
Neon_Hills(config)#exit
Neon_Hills#
```

For Solar_Palace:

```
Solar_Palace>en
Solar_Palace#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Solar_Palace(config)#hostname Solar_Palace
Solar_Palace(config)#interface f0/0
Solar_Palace(config-if)#ip address 18.69.24.1 255.255.255.192
Solar_Palace(config-if)#no shut
Solar_Palace(config-if)#exit
```

Solar_Palace(config)#exit
Solar_Palace#

Communication between Solar & Neon
On Solar Palace
Solar_Palace#conf t
Solar_Palace(config)#interface serial 0/0/0
Solar_Palace(config-if)#ip address 18.69.24.65 255.255.255.252
Solar_Palace(config-if)#clock rate 64000
Solar_Palace(config-if)#no shutdown
Solar_Palace(config-if)#exit
Solar_Palace(config)#exit

On Neon Hills
Neon_Hills#conf t
Neon_Hills(config)#interface serial 0/0/0
Neon_Hills(config-if)#ip address 18.69.24.66 255.255.255.252
Neon_Hills(config-if)#no shutdown
Neon_Hills(config-if)#exit
Neon_Hills(config)#exit

Solar palace Ip route static
ip route 18.69.23.0 255.255.255.0 18.69.24.66
Neon Hills Ip route static
ip route 18.69.24.0 255.255.255.192 18.69.24.65

Communication between Neon and Titan
On Neon Hills
Neon_Hills#conf t
Neon_Hills(config)#interface serial 0/0/1
Neon_Hills(config-if)#ip address 18.69.24.69 255.255.255.252
Neon_Hills(config-if)#clock rate 64000
Neon_Hills(config-if)#no shutdown
Neon_Hills(config-if)#exit
Neon_Hills(config)#exit

Titan Valley
Titan_Valley#conf t
Titan_Valley(config)#interface serial 0/0/1
Titan_Valley(config-if)#ip address 18.69.24.70 255.255.255.252
Titan_Valley(config-if)#no shutdown
Titan_Valley(config-if)#exit
Titan_Valley(config)#exit

Neon Hills ip route static
ip route 18.69.0.0 255.255.240.0 serial 0/0/1
Titan Valley ip route static
ip route 18.69.23.0 255.255.255.0 serial 0/0/0

Solar_Palace> enable
Solar_Palace# conf t
ip route 18.69.23.0 255.255.255.0 18.69.24.66
ip route 18.69.0.0 255.255.240.0 18.69.24.66
exit

Neon_Hills> enable
Neon_Hills# conf t
ip route 18.69.24.0 255.255.255.192 18.69.24.65
ip route 18.69.0.0 255.255.240.0 18.69.24.70
exit

Titan_Valley> enable
Titan_Valley# conf t
ip route 18.69.23.0 255.255.255.0 18.69.24.69
ip route 18.69.24.0 255.255.255.192 18.69.24.69
exit

Static Floating Routing
enable
configure terminal
ip route 18.69.23.0 255.255.255.0 18.69.24.73 150

```
exit
write memory

Titan_Valley>en
Titan_Valley#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Titan_Valley(config)#router rip
Titan_Valley(config-router)#version 2
Titan_Valley(config-router)#no auto-summary
Titan_Valley(config-router)#network 18.69.0.0
Titan_Valley(config-router)#network 18.69.20.0
Titan_Valley(config-router)#network 18.69.22.0
Titan_Valley(config-router)#exit
Titan_Valley(config)#exit


Aurora_Ridge#en
Aurora_Ridge#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Aurora_Ridge(config)#interface serial 0/0/0
Aurora_Ridge(config-if)#ip address 18.69.30.1 255.255.255.252
Aurora_Ridge(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Aurora_Ridge(config-if)#exit
Aurora_Ridge(config)#exit
Aurora_Ridge#
%SYS-5-CONFIG_I: Configured from console by console

Aurora_Ridge#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Aurora_Ridge(config)#router rip
Aurora_Ridge(config-router)#version 2
Aurora_Ridge(config-router)#no auto-summary
Aurora_Ridge(config-router)#network 18.69.22.0
Aurora_Ridge(config-router)#network 18.69.0.0
Aurora_Ridge(config-router)#exit
```

```
Aurora_Ridge(config)#exit
```

```
Glacier_Fields> enable
Glacier_Fields# conf t
interface serial 0/0/1
ip address 18.69.30.2 255.255.255.252
no shutdown
exit
```

```
Glacier_Fields>en
Glacier_Fields#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Glacier_Fields(config)#router rip
Glacier_Fields(config-router)#version 2
Glacier_Fields(config-router)#no auto-summary
Glacier_Fields(config-router)#network 18.69.20.0
Glacier_Fields(config-router)#network 18.69.0.0
Glacier_Fields(config-router)#exit
Glacier_Fields(config)#exit
Glacier_Fields#
```

```
Neon_Hills#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Neon_Hills(config)#interface serial 0/1/0
Neon_Hills(config-if)#ip address 18.69.30.5 255.255.255.252
Neon_Hills(config-if)#clock rate 64000
Neon_Hills(config-if)#no shut
```

```
Glacier_Fields#
Glacier_Fields#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Glacier_Fields(config)#interface serial 0/0/0
```

Glacier_Fields(config-if)#ip address 18.69.30.6 255.255.255.252
Glacier_Fields(config-if)#no shut
Glacier_Fields(config-if)#exit
Glacier_Fields(config)#exit

DHCP server routing

DHCP Server Configuration:
hostname Neon_Hills

interface Serial0/1/0
ip address 18.69.30.5 255.255.255.252
no shutdown

router rip
version 2
network 18.0.0.0
no auto-summary

ip dhcp pool GLACIER_FIELDS
network 18.69.20.0 255.255.254.0
default-router 18.69.20.1
dns-server 18.69.23.2

ip dhcp pool AURORA_RIDGE
network 18.69.22.0 255.255.255.0
default-router 18.69.22.1
dns-server 18.69.23.2

hostname Glacier_Fields

interface FastEthernet0/1
ip address 18.69.20.1 255.255.254.0
ip helper-address 18.69.23.10
no shutdown

```
duplex auto
speed auto

interface Serial0/0/0
ip address 18.69.30.6 255.255.255.252
no shutdown

interface Serial0/0/1
ip address 18.69.30.2 255.255.255.252
clock rate 64000
no shutdown

router rip
version 2
network 18.0.0.0
no auto-summary


hostname Aurora_Ridge

interface FastEthernet0/0
ip address 18.69.22.1 255.255.255.0
ip helper-address 18.69.23.10
no shutdown
duplex auto
speed auto

interface Serial0/0/0
ip address 18.69.30.1 255.255.255.252
no shutdown

router rip
version 2
network 18.0.0.0
no auto-summary
```

Assumption: Distance less than 200 are considered to be directly neighbours.